

Supporting Information

**Heavy-Atom Tunneling Calculations in Thirteen Organic Reactions:
Tunneling Contributions are Substantial, and Bell's Formula Closely
Approximates Multidimensional Tunneling at ≥ 250 K**

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SUPPORTING INFORMATION**Table of Contents**

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Bell^{5,6} and Truhlar⁷ functions for the transmission coefficient

Equations contain temperature T , Boltzmann constant k_B , Planck's constant h , imaginary frequency $iv^\ddagger$, barrier height V_0 .

Bell formula:

$$\kappa(T) = \frac{u/2}{\sin(u/2)} - \sum_{n=1}^{\infty} (-1)^n \frac{\exp\left[\left(\frac{u-2n\pi}{u}\right)\left(\frac{V_0}{k_B T}\right)\right]}{\frac{u-2n\pi}{u}} \quad (S1)$$

$$u = hv^\ddagger/k_B T$$

Truhlar formulas from Ref. 7:
(assume exoergic reaction)

$$\kappa(T) \cong \frac{\pi\beta/\alpha}{\sin(\pi\beta/\alpha)} + \frac{\beta}{\beta-\alpha} e^{(\beta-\alpha)V_0} \quad (\alpha > \beta) \quad (S2)$$

$$\kappa(T) = \alpha V_0 = \beta V_0 \quad (\alpha = \beta)$$

$$\kappa(T) \cong \frac{\beta}{\beta-\alpha} \{\exp[(\beta-\alpha)V_0] - 1\} \quad (\beta > \alpha) \quad (S3)$$

$$\alpha = 2\pi/hv^\ddagger$$

$$\beta = 1/k_B T$$

Connection with Bell: $u = 2\pi\beta/\alpha$

Typical conditions for
use of S2 and S3:

S2: small $v^\ddagger$, high T
(reactions R1-R13)

S3: large $v^\ddagger$, low T

Transmission coefficient tables for κ_{SCT} , κ_{Bell} , κ_{W} **Table S1. κ_{SCT} from Polyrate**

Temp, K	450	400	350	300	250
R1	1.29	1.38	1.54	1.84	2.57
R2	1.16	1.21	1.28	1.42	1.68
R3	1.15	1.20	1.27	1.39	1.63
R4	1.11	1.15	1.20	1.28	1.44
R5	1.12	1.15	1.21	1.29	1.46
R6	1.10	1.13	1.18	1.25	1.39
R7	1.24	1.31	1.43	1.66	2.16
R8	1.17	1.23	1.32	1.47	1.78
R9	1.14	1.16	1.21	1.31	1.48
R10	1.17	1.23	1.32	1.47	1.78
R11	1.10	1.12	1.16	1.23	1.36
R12	1.20	1.26	1.36	1.53	1.90
R13	1.13	1.16	1.22	1.31	1.50

Table S2. Wigner κ_{W} and unsigned errors vs κ_{SCT}

T (K)	---- κ_{W} ----			Error: $ \kappa_{\text{SCT}} - \kappa_{\text{W}} $		
	400	300	250	400	300	250
R1	1.30	1.53	1.76	0.08	0.31	0.80
R2	1.18	1.31	1.45	0.03	0.10	0.23
R3	1.17	1.31	1.45	0.02	0.08	0.18
R4	1.13	1.22	1.32	0.02	0.06	0.12
R5	1.13	1.24	1.34	0.02	0.06	0.12
R6	1.11	1.20	1.29	0.02	0.05	0.10
R7	1.27	1.49	1.70	0.04	0.17	0.46
R8	1.22	1.38	1.55	0.02	0.09	0.23
R9	1.22	1.39	1.55	0.06	0.08	0.07
R10	1.17	1.31	1.45	0.06	0.16	0.33
R11	1.10	1.18	1.26	0.02	0.05	0.09
R12	1.21	1.37	1.53	0.05	0.16	0.37
R13	1.14	1.24	1.35	0.03	0.07	0.15
				0.04	0.11	0.25

← MUE

Table S3. Bell κ_{Bell} and unsigned errors vs κ_{SCT}

T (K)	---- κ_{Bell} ----			Error: $ \kappa_{\text{SCT}} - \kappa_{\text{Bell}} $		
	400	300	250	400	300	250
R1	1.37	1.82	2.53	0.01	0.02	0.03
R2	1.20	1.40	1.65	0.01	0.02	0.03
R3	1.20	1.40	1.64	0.00	0.01	0.02
R4	1.14	1.26	1.41	0.01	0.02	0.03
R5	1.15	1.28	1.44	0.01	0.01	0.02
R6	1.12	1.23	1.36	0.01	0.02	0.03
R7	1.34	1.72	2.31	0.03	0.07	0.15
R8	1.25	1.52	1.88	0.02	0.05	0.10
R9	1.26	1.52	1.89	0.10	0.22	0.41
R10	1.20	1.39	1.64	0.03	0.08	0.13
R11	1.11	1.21	1.32	0.01	0.02	0.04
R12	1.24	1.49	1.82	0.02	0.04	0.07
R13	1.15	1.29	1.46	0.01	0.02	0.04
				0.02	0.04	0.08

← MUE

← excluding R9

S4

0.01

0.03

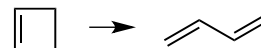
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Electronic structure calculations

Stationary points for most reactions were taken from the literature (ref 8). Reactant and product geometries were used as given. Saddle points were reoptimized such that the largest absolute gradient component was < 0.00001 hartree/bohr. Saddle point geometries (Å) and energies are listed below. Important Gaussian 09 route directives are listed (excluding those required by Gaussrate), plus reaction energetics written by Polyrate.

R1

b3lyp/6-31g* int=ultra
E = -155.916553896674 au



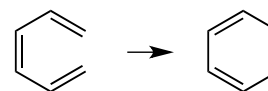
	X	Y	Z
h	1.882085075	-1.124576587	-0.435334902
c	1.064866756	-0.649843000	0.119100620
c	-1.065442200	-0.648901334	-0.119087898
c	0.683671360	0.709194810	-0.083819054
c	-0.683042613	0.709802104	0.083805164
h	-1.883081045	-1.122899487	0.435356837
h	-1.338894524	1.518544630	0.403250228
h	0.871053929	-1.110498116	1.080531468
h	-0.872038018	-1.109746864	-1.080509782
h	1.340239883	1.517349447	-0.403279910

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.05671	1.54321	12446.71	35.5869
V w/re product V	0.06993	1.90295	15348.22	43.8828
V+ZPE w/re reactant V	0.14093	3.83489	30930.23	88.4340
V+ZPE w/re product V	0.15415	4.19464	33831.75	96.7298
V+ZPE w/re reactant V+ZPE	0.05400	1.46934	11850.97	33.8836
V+ZPE w/re product V+ZPE	0.06881	1.87238	15101.63	43.1777
V+ZPE w/re saddle point V	0.08422	2.29168	18483.53	52.8471

R2

b3lyp/6-31g* int=ultra
E = -233.347290804755 au



	X	Y	Z
c	-1.227241998	1.140778582	-0.102618891
c	0.096244731	1.482611791	0.192269731
c	1.218752510	0.702978133	-0.102655859
c	1.218752510	-0.702978133	-0.102655859
c	0.096244731	-1.482611791	0.192269731
c	-1.227241998	-1.140778582	-0.102618891
h	0.272735872	2.340219928	0.845334520
h	2.196474842	1.175092134	-0.013240429
h	2.196474842	-1.175092134	-0.013240429
h	0.272735872	-2.340219928	0.845334520
h	-2.022789647	-1.641165903	0.450543743
h	-1.491307753	-0.936629598	-1.127789279
h	-2.022789647	1.641165903	0.450543743
h	-1.491307753	0.936629598	-1.127789279

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm ⁻¹	kcal
V w/re reactants V	0.03227	0.87808	7082.10	20.2487
V w/re product V	0.07163	1.94904	15719.90	44.9455
V+ZPE w/re reactant V	0.15163	4.12612	33279.09	95.1497
V+ZPE w/re product V	0.19099	5.19708	41916.89	119.8464
V+ZPE w/re reactant V+ZPE	0.03269	0.88949	7174.19	20.5120
V+ZPE w/re product V+ZPE	0.06819	1.85553	14965.74	42.7892
V+ZPE w/re saddle point V	0.11936	3.24804	26196.99	74.9010

R3

b3lyp/6-31g* int=ultra

E = -234.556931055085 au



	X	Y	Z
c	1.431738763	0.000460490	-0.260244567
c	-1.431738435	-0.000460485	0.260244763
c	0.950419567	-1.218276013	0.253966357
c	-0.950419741	1.218275880	-0.253966442
c	0.949639210	1.218884154	0.253976348
c	-0.949639385	-1.218884021	-0.253976417
h	1.312013333	-2.144838646	-0.191697987
h	-1.312013397	2.144838609	0.191697712
h	0.814639030	-1.299807260	1.330982819
h	-0.814639104	1.299806979	-1.330982909
h	0.813805608	1.300319927	1.330993275
h	-0.813805698	-1.300319673	-1.330993349
h	1.310637827	2.145682106	-0.191681154
h	-1.310637891	-2.145682075	0.191680916
h	1.823868842	0.000590329	-1.277825555
h	-1.823868334	-0.000590324	1.277825735

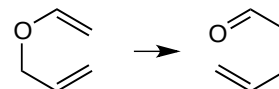
Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm ⁻¹	kcal
V w/re reactants V	0.05377	1.46321	11801.46	33.7421
V w/re product V	0.05377	1.46321	11801.46	33.7421
V+ZPE w/re reactant V	0.19579	5.32786	42971.71	122.8623
V+ZPE w/re product V	0.19579	5.32786	42971.71	122.8623
V+ZPE w/re reactant V+ZPE	0.05334	1.45138	11706.07	33.4693
V+ZPE w/re product V+ZPE	0.05334	1.45138	11706.07	33.4693
V+ZPE w/re saddle point V	0.14202	3.86465	31170.25	89.1202

R4

b3lyp/6-31g* int=ultra

E = -270.461121759637 au



	X	Y	Z
o	-0.543109741	-1.303767590	-0.236431920
c	-1.294540059	-0.394099369	0.279655789
c	-1.435207217	0.865943936	-0.273148185
c	0.721162044	1.444880850	0.328719179
c	1.393932171	0.411231600	-0.297437700
c	1.268552012	-0.897898607	0.188776257
h	-1.193785929	1.027132021	-1.317380686

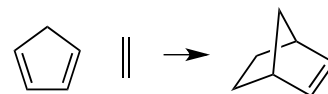
h	-2.091679689	1.594407534	0.197572393
h	0.708516013	2.444806611	-0.096204757
h	0.501307200	1.395180171	1.390687363
h	1.710820960	0.544781448	-1.330045455
h	1.722630389	-1.724078630	-0.347692887
h	1.156813035	-1.062421757	1.256339245
h	-1.680941547	-0.555535951	1.301395257

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.04405	1.19876	9668.60	27.6439
V w/re product V	0.07231	1.96766	15870.09	45.3749
V+ZPE w/re reactant V	0.16111	4.38394	35358.60	101.0953
V+ZPE w/re product V	0.18936	5.15284	41560.09	118.8263
V+ZPE w/re reactant V+ZPE	0.04215	1.14688	9250.11	26.4474
V+ZPE w/re product V+ZPE	0.07097	1.93124	15576.40	44.5352
V+ZPE w/re saddle point V	0.11705	3.18518	25690.00	73.4514

R5

b3lyp/6-31g* int=ultra
E = -272.656770583146 au



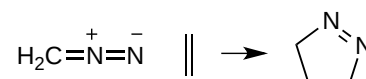
	X	Y	Z
c	-1.073216942	0.563564143	-0.438479267
c	-1.073281956	0.563951108	0.959602932
c	0.256141270	0.563953594	1.416743017
c	1.115920637	0.563568233	0.314285179
c	0.299015550	1.027247805	-0.869581967
c	0.864784622	-1.560433948	-0.380696370
c	-0.447765008	-1.560436410	-0.832034244
h	0.562262111	0.443563510	2.451242145
h	0.635892978	0.682530348	-1.849263768
h	0.302238137	2.131118948	-0.878959705
h	2.192083375	0.691896602	0.376848183
h	-1.950888833	0.443558808	1.587061337
h	-1.960271512	0.691888831	-1.050994985
h	-0.661069476	-1.480093272	-1.894226455
h	1.120687049	-2.023152831	0.566025251
h	1.686235376	-1.480088864	-1.087074055
h	-1.231788637	-2.023157246	-0.242905209

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.03175	0.86399	6968.49	19.9239
V w/re product V	0.07063	1.92183	15500.51	44.3182
V+ZPE w/re reactant V	0.17969	4.88966	39437.47	112.7574
V+ZPE w/re product V	0.21857	5.94751	47969.49	137.1517
V+ZPE w/re reactant V+ZPE	0.03559	0.96843	7810.85	22.3323
V+ZPE w/re product V+ZPE	0.06495	1.76738	14254.77	40.7564
V+ZPE w/re saddle point V	0.14794	4.02567	32468.98	92.8335

R6

b3lyp/6-31g* int=ultra



E = -227.303985943672 au

	X	Y	Z
c	-1.336095087	0.800283927	0.000211216
c	-1.497198446	-0.564108901	0.000248454
h	-1.470475225	1.368412790	0.916628042
h	-1.670849515	-1.113231363	0.920206830
h	-1.671149268	-1.113246080	-0.919644526
h	-1.470774385	1.368398100	-0.916170801
c	0.895242879	1.127013559	-0.000155663
h	1.188555395	1.636019894	0.916581168
h	1.188255896	1.636003642	-0.916997281
n	1.194841426	-0.185769410	-0.000193033
n	0.747133638	-1.254644508	-0.000110916

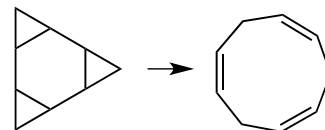
Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.02273	0.61856	4989.01	14.2643
V w/re product V	0.08176	2.22488	17944.70	51.3065
V+ZPE w/re reactant V	0.10968	2.98470	24073.07	68.8284
V+ZPE w/re product V	0.16872	4.59102	37028.76	105.8706
V+ZPE w/re reactant V+ZPE	0.02650	0.72114	5816.31	16.6297
V+ZPE w/re product V+ZPE	0.07533	2.04997	16533.98	47.2730
V+ZPE w/re saddle point V	0.08695	2.36614	19084.06	54.5641

R7

b3lyp/6-31g* int=ultra

E = -350.042416349683 au



	X	Y	Z
h	1.181257309	2.077060085	-1.132558754
c	0.707164701	1.486224924	-0.355411000
c	0.000000104	-1.858865980	0.691083591
c	1.609825109	0.929433081	0.691083591
h	2.389415455	-0.015530936	-1.132558754
h	1.139859438	0.658098239	1.637060172
h	2.543218103	1.468327847	0.863079187
c	1.640690905	-0.130689682	-0.355411000
c	0.933526190	-1.355535058	-0.355411000
h	-2.389415453	-0.015531204	-1.132558754
c	-1.640690891	-0.130689866	-0.355411000
h	0.000000165	-2.936655408	0.863079187
c	-1.609825213	0.929432900	0.691083591
h	-1.181257543	2.077059952	-1.132558754
h	-1.139859512	0.658098111	1.637060172
h	-2.543218268	1.468327561	0.863079187
c	-0.707164868	1.486224845	-0.355411000
h	1.208158144	-2.061528881	-1.132558754
h	0.000000074	-1.316196349	1.637060172
c	-0.933526037	-1.355535163	-0.355411000
h	-1.208157912	-2.061529016	-1.132558754

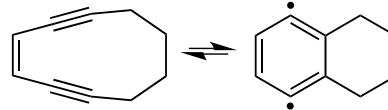
Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.03842	1.04542	8431.79	24.1077
V w/re product V	0.07012	1.90816	15390.25	44.0029
V+ZPE w/re reactant V	0.22233	6.05000	48796.13	139.5152

V+ZPE w/re product V	0.25404	6.91275	55754.60	159.4104
V+ZPE w/re reactant V+ZPE	0.03480	0.94696	7637.68	21.8372
V+ZPE w/re product V+ZPE	0.06788	1.84722	14898.70	42.5975
V+ZPE w/re saddle point V	0.18391	5.00458	40364.35	115.4075

R8

blyp/6-31g* int=ultra iop(3/76=10000000000)
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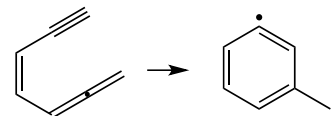
	X	Y	Z
c	0.013803893	-0.690064055	2.643131274
c	-0.013803893	0.690064055	2.643131274
c	0.026595869	-1.380906086	1.409144996
c	-0.026595869	1.380906086	1.409144996
c	0.010828463	-0.950619328	0.189969234
c	-0.010828463	0.950619328	0.189969234
c	-0.078505712	-1.571411730	-1.177244472
c	0.078505712	1.571411730	-1.177244472
c	0.368708835	-0.668406990	-2.325372017
c	-0.368708835	0.668406990	-2.325372017
h	0.021887438	-1.219509869	3.577193059
h	-0.021887438	1.219509869	3.577193059
h	0.510644688	-2.481638580	-1.174011167
h	-0.510644688	2.481638580	-1.174011167
h	-1.109985854	-1.869035237	-1.346949755
h	1.109985854	1.869035237	-1.346949755
h	1.440605966	-0.493549968	-2.264861497
h	-1.440605966	0.493549968	-2.264861497
h	0.189358446	-1.191871607	-3.259896955
h	-0.189358446	1.191871607	-3.259896955

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm ⁻¹	kcal
V w/re reactants V	0.04068	1.10701	8928.60	25.5282
V w/re product V	0.02952	0.80332	6479.15	18.5248
V+ZPE w/re reactant V	0.20076	5.46305	44062.08	125.9798
V+ZPE w/re product V	0.18960	5.15935	41612.63	118.9765
V+ZPE w/re reactant V+ZPE	0.03969	1.07990	8709.89	24.9028
V+ZPE w/re product V+ZPE	0.02699	0.73435	5922.86	16.9343
V+ZPE w/re saddle point V	0.16008	4.35603	35133.47	100.4517

R9

blyp/6-31g* int=ultra iop(3/76=10000000000)
iop(3/77=1422210319) iop(3/78=1185605343)
E = -270.393800820421 au



	X	Y	Z
c	2.05974241	-0.16778537	-2.36408785
h	3.04593773	-0.64598737	-4.07589931
c	3.24890263	1.27696484	-0.58864846
h	5.16713416	1.88775901	-0.84073915
c	-0.35876324	-1.34510347	-1.96719900
h	-0.90633482	-2.76193488	-3.31988772
c	-1.89277108	-1.05442614	0.07899225
c	-4.35296682	-1.48240027	0.73236671
h	-5.87256533	-0.66883147	-0.34651257

S10

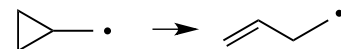
h	-4.83023949	-2.24773878	2.55318933
c	1.92428618	1.81836500	1.68303233
c	-0.21744776	1.18857571	2.59046543
h	-1.49742830	1.64826954	4.06615832

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm ⁻¹	kcal
V w/re reactants V	0.03755	1.02177	8241.07	23.5624
V w/re product V	0.05434	1.47875	11926.84	34.1006
V+ZPE w/re reactant V	0.13080	3.55924	28706.94	82.0773
V+ZPE w/re product V	0.14759	4.01622	32392.71	92.6154
V+ZPE w/re reactant V+ZPE	0.03827	1.04135	8398.95	24.0138
V+ZPE w/re product V+ZPE	0.05126	1.39481	11249.79	32.1648
V+ZPE w/re saddle point V	0.09325	2.53747	20465.87	58.5149

R10

ub3lyp/6-31g* int=ultra
E = -156.531038331252 au



	X	Y	Z
c	1.604145234	-0.123201319	-0.187349940
c	0.418782348	0.138535406	0.478440663
c	-0.806692531	0.725238713	-0.147569051
c	-1.194572726	-0.705261881	-0.118914889
h	1.712604102	0.075606348	-1.250311209
h	2.438660939	-0.598800820	0.317305728
h	0.386517775	0.009066730	1.557265964
h	-1.751703511	-1.118828125	0.713523934
h	-1.008109422	-1.339567193	-0.975992039
h	-1.387247304	1.399520408	0.484841584
h	-0.648652192	1.152561583	-1.139622759

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm ⁻¹	kcal
V w/re reactants V	0.01332	0.36251	2923.81	8.3596
V w/re product V	0.01695	0.46130	3720.57	10.6377
V+ZPE w/re reactant V	0.10711	2.91464	23508.00	67.2127
V+ZPE w/re product V	0.11074	3.01343	24304.76	69.4908
V+ZPE w/re reactant V+ZPE	0.01218	0.33147	2673.48	7.6439
V+ZPE w/re product V+ZPE	0.01705	0.46405	3742.77	10.7011
V+ZPE w/re saddle point V	0.09379	2.55213	20584.18	58.8531

R11

uhcth/6-31+g** int=ultra
E = -235.190357843256 au



	X	Y	Z
c	-1.669327411	0.393733163	-0.023763216
c	-1.060292412	-0.931881837	0.444475786
h	-2.522753409	0.659288165	0.621327783
h	-2.071196413	0.275741164	-1.038737214
h	-1.776063412	-1.760017836	0.360205783
h	-0.792581410	-0.849386836	1.506821786
c	1.095958589	0.027002166	-0.472121215

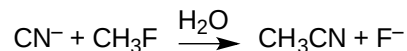
c	2.135375589	0.256706166	0.396604785
c	0.206030590	-1.199767837	-0.377058216
h	-0.082619410	-1.512533835	-1.389019216
h	0.766205588	-2.035940836	0.058099783
c	-0.616301411	1.450576166	-0.000604214
h	-0.289740412	1.822649163	0.968283786
h	-0.563510412	2.180678165	-0.805947217
h	2.841720589	1.066541163	0.233522785
h	2.239759587	-0.305643835	1.322969786
h	1.161976591	0.501871165	-1.450957216

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.01241	0.33774	2724.06	7.7885
V w/re product V	0.03617	0.98417	7937.82	22.6954
V+ZPE w/re reactant V	0.16260	4.42470	35687.33	102.0352
V+ZPE w/re product V	0.18636	5.07113	40901.09	116.9421
V+ZPE w/re reactant V+ZPE	0.01333	0.36282	2926.29	8.3667
V+ZPE w/re product V+ZPE	0.03388	0.92192	7435.71	21.2598
V+ZPE w/re saddle point V	0.15019	4.08696	32963.27	94.2467

R12

b3lyp/6-31++g(d,p) int=ultra scrf(pcm,solvent=water)
E = -232.697450999917 au



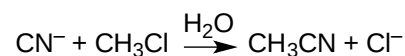
	X	Y	Z
c	0.000000000	0.000000000	0.619220743
f	0.000000000	0.000000000	2.451409742
h	0.000000000	1.075622001	0.583569743
h	0.931516001	-0.537811000	0.583569743
h	-0.931516001	-0.537811000	0.583569743
c	0.000000000	0.000000000	-1.511642261
n	0.000000000	0.000000000	-2.687140259

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.03182	0.86588	6983.77	19.9676
V w/re product V	0.07221	1.96499	15848.55	45.3133
V+ZPE w/re reactant V	0.07618	2.07310	16720.58	47.8065
V+ZPE w/re product V	0.11658	3.17221	25585.35	73.1522
V+ZPE w/re reactant V+ZPE	0.03216	0.87521	7059.01	20.1827
V+ZPE w/re product V+ZPE	0.07134	1.94118	15656.53	44.7643
V+ZPE w/re saddle point V	0.04436	1.20722	9736.80	27.8389

R13

b3lyp/6-31++g(d,p) int=ultra scrf(pcm,solvent=water)
E = -593.070160630155 au



	X	Y	Z
c	0.012336801	-0.000079392	-0.000410779
cl	-2.202433197	0.000043604	0.000148217
h	0.102107801	-0.501342395	-0.948262784
h	0.102622802	1.071384604	0.039384217
h	0.102617801	-0.570334394	0.907540219

c	2.310408805	-0.000265393	-0.000466782
n	3.487367802	0.000207607	0.000478217

Saddle point energetics (V = classical energy, ZPE = zero point energy)

	hartrees	eV	cm** ⁻¹	kcal
V w/re reactants V	0.01888	0.51376	4143.71	11.8475
V w/re product V	0.08923	2.42804	19583.32	55.9915
V+ZPE w/re reactant V	0.06205	1.68836	13617.45	38.9343
V+ZPE w/re product V	0.13239	3.60265	29057.06	83.0783
V+ZPE w/re reactant V+ZPE	0.01939	0.52760	4255.35	12.1667
V+ZPE w/re product V+ZPE	0.08715	2.37162	19128.24	54.6904
V+ZPE w/re saddle point V	0.04317	1.17460	9473.75	27.0868

Procedure for Polyrate calculations

Reaction paths were computed by the Page-McIver method with cubic first step. Second derivatives were computed at every step (*inh*=1). The initial step size (*fsize*) was 0.001 Å-amu^{1/2} within 1.5 kcal/mol of the saddle point in either direction. Thereafter, the step size (*sstep*) was 0.01 Å-amu^{1/2}. The length of the MEP (IRC) was determined by *percentdown*, which was set to a value such that the transmission probability at the MEP termini was zero, as printed in the "Transmission probability" section of the Polyrate long output (*.fu6).

Polyrate dat file for reaction R2 (other dat files are similar)

*General

TITLE

Cis-hexa-1,3,5-triene cyclization

END

ATOMS

1	C
2	C
3	C
4	C
5	C
6	C
7	H
8	H
9	H
10	H
11	H
12	H
13	H
14	H

END

NOSUPERMOL

writefu31

*SECOND

HESSCAL HHOOK

*OPTIMIZATION

OPTTS OHOOK

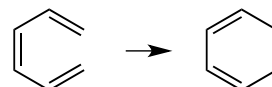
OPTMIN OHOOK

*REACT1

status 2

GEOM

1	-1.169096	1.626489	0.427095
2	0.064923	1.645713	-0.092676
3	1.140385	0.661760	0.102493
4	1.110517	-0.679608	0.256107
5	0.007106	-1.647686	0.165687



S14

6	-1.050161	-1.612286	-0.655442
7	0.362057	2.523893	-0.668603
8	2.132988	1.113245	0.139326
9	2.072510	-1.145292	0.475585
10	0.132587	-2.528458	0.797891
11	-1.777525	-2.419286	-0.665763
12	-1.223361	-0.785750	-1.337698
13	-1.862774	2.443978	0.251209
14	-1.524619	0.803288	1.039116

END

SPECIES NONLINRP

*PROD1

status 2

GEOM

1	-1.153117	0.746792	-0.289443
2	0.108619	1.417123	0.205776
3	1.248880	0.717055	0.324859
4	1.283078	-0.717474	0.018562
5	0.144614	-1.417196	-0.118409
6	-1.195361	-0.746468	0.084459
7	0.089226	2.487449	0.399536
8	2.171195	1.205105	0.632030
9	2.250416	-1.205813	-0.076273
10	0.168386	-2.487523	-0.311675
11	-1.482037	-0.853392	1.144839
12	-1.977517	-1.254249	-0.492074
13	-2.043568	1.254828	0.099139
14	-1.197275	0.853764	-1.386999

END

SPECIES NONLINRP

*START

status 2

GEOM

1	-1.2036847934	1.1407785812	-0.1061445365
2	0.1198019358	1.482611789	0.1887440816
3	1.2423097171	0.7029781345	-0.1061815088
4	1.2423097171	-0.7029781345	-0.1061815088
5	0.1198019358	-1.482611789	0.1887440816
6	-1.2036847934	-1.1407785812	-0.1061445365
7	0.2962930765	2.3402199276	0.8418088743
8	2.220032048	1.175092135	-0.0167660763
9	2.220032048	-1.175092135	-0.0167660763
10	0.2962930765	-2.3402199276	0.8418088743
11	-1.9992324395	-1.6411659039	0.4470180966
12	-1.4677505463	-0.9366295973	-1.1313149244
13	-1.9992324395	1.6411659039	0.4470180966
14	-1.4677505463	0.9366295973	-1.1313149244

END

SPECIES NONLINTS

*PATH

dlx3	0.001
SCALEMASS	1.0
SSTEP	0.01
INH	1
NSTEPS	99999
CURV	dhess
RPM	pagem
FIRSTSTEP	cubic
SIGN	product

SRANGE

slp	8.
slm	-8.

END

prsavevp

specstop

percentdown	94.
-------------	-----

end

sfirst

nfstep	340
fsize	0.001

end

*TUNNEL

ZCT

SCT

*RATE

forwardk

SIGMAF	1
--------	---

TST

CVT

TEMP

200.
250.
300.
350.
400.
450.
500.

END

For each reaction R1-R13, the following information is given:

- transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$
- plots of $\kappa_{\text{SCT}}(T)$ contributions as a function of V_{a}^{G} energy
- rate constants

TST: transition state theory (conventional)

CVT: canonical variational transition state theory

ZCT: zero-curvature tunneling

SCT: small-curvature tunneling

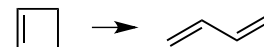
ZPE: zero-point vibrational energy

V_{a}^{G} : vibrationally adiabatic ground state energy (IRC or MEP + ZPE)

Note on $\kappa_{\text{SCT}}(T)$ plots:

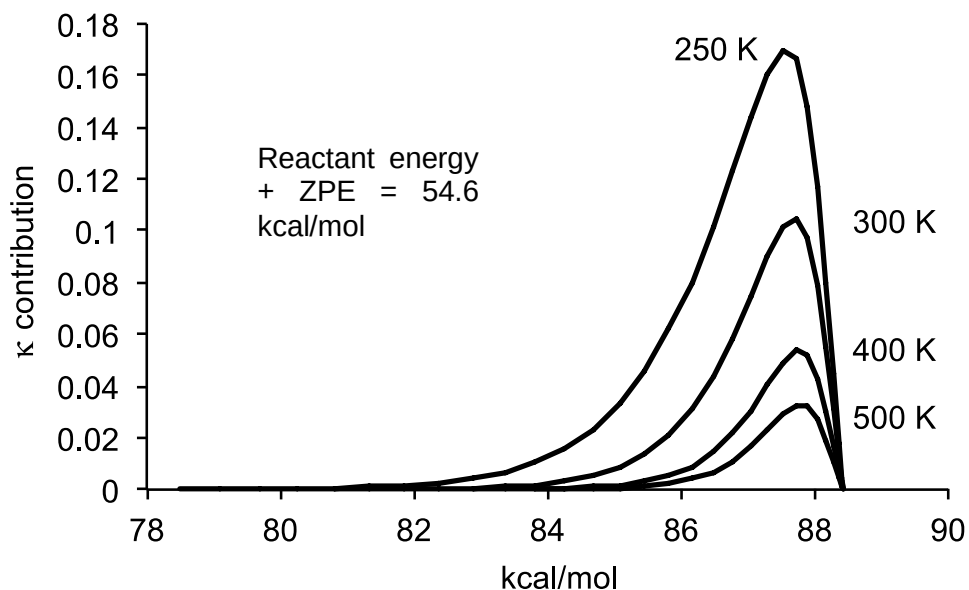
These plots show the *net* contributions to the Boltzmann-weighted integral of transmission probability over energy that gives $\kappa_{\text{SCT}}(T)$, as a function of V_{a}^{G} energy at selected temperatures T . The integral includes transmission through the barrier and reflection above the barrier as described in ref 3a (the procedure used in Polyrate). Although the integration is formally from the ZPE-corrected reactant energy E_{R} to infinity, the plots stop at the V_{a}^{G} maximum, $V_{\text{a}}^{\text{G}}_{\text{MAX}}$, where transmission and reflection sum to zero. The integration scheme in ref 3a includes both the through-barrier and above-barrier contributions in a single integration from E_{R} to $V_{\text{a}}^{\text{G}}_{\text{MAX}}$, by assuming that transmission and reflection probabilities are symmetrical about $V_{\text{a}}^{\text{G}}_{\text{MAX}}$ up to $2V_{\text{a}}^{\text{G}}_{\text{MAX}} - E_{\text{R}}$. For higher energies the motion is assumed to be classical. Tunneling contributions to κ_{SCT} are larger than reflection contributions because of the larger Boltzmann factors.

In the $\kappa_{\text{SCT}}(T)$ plots, the Polyrate zero of energy is the classical reactant energy (without ZPE). The V_{a}^{G} energies on the horizontal axis include ZPE, and the ZPE-corrected reactant energy is shown on the plot for comparison.

R1Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
250.00	2.4168E+00	2.4690E+00
300.00	1.7791E+00	1.7972E+00
350.00	1.5062E+00	1.5153E+00
400.00	1.3600E+00	1.3654E+00
450.00	1.2710E+00	1.2747E+00
500.00	1.2123E+00	1.2149E+00

$$v_{\text{imag}}^{\ddagger} = 742.4 \text{ cm}^{-1}$$

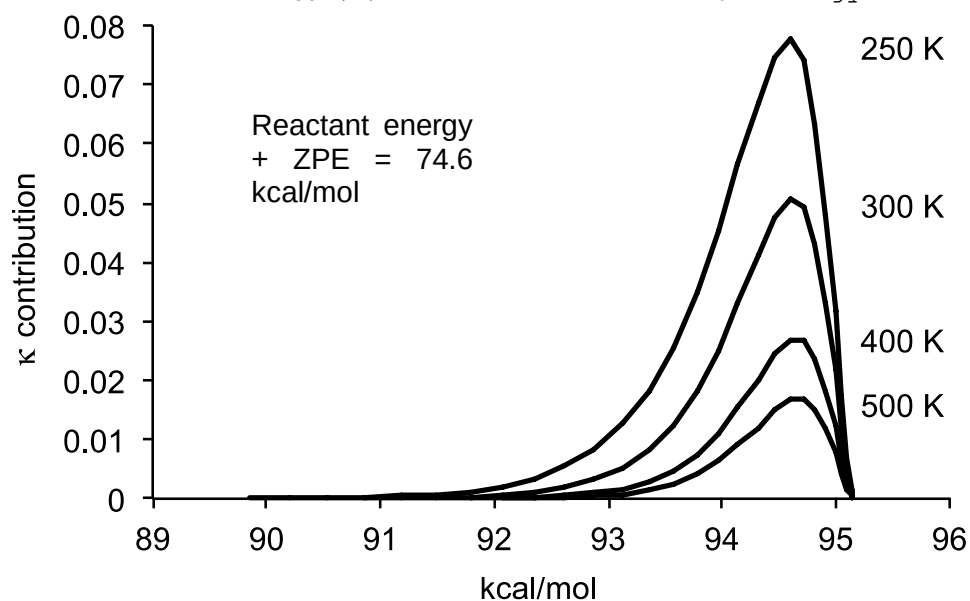
Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_a^{G} energy at 250-500 K.Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
250.00	1.33E-17	1.33E-17	3.20E-17	3.41E-17
300.00	1.36E-12	1.36E-12	2.42E-12	2.51E-12
350.00	5.30E-09	5.30E-09	7.97E-09	8.17E-09
400.00	2.66E-06	2.66E-06	3.61E-06	3.68E-06
450.00	3.40E-04	3.40E-04	4.32E-04	4.37E-04
500.00	1.66E-02	1.66E-02	2.01E-02	2.04E-02

R2Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
250.00	1.6307E+00	1.6782E+00
300.00	1.3897E+00	1.4152E+00
350.00	1.2681E+00	1.2843E+00
400.00	1.1970E+00	1.2083E+00
450.00	1.1515E+00	1.1599E+00
500.00	1.1204E+00	1.1269E+00

$$v_{\text{imag}}^{\ddagger} = 571.8 \text{ cm}^{-1}$$

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_a^{G} energy at 250–500 K.Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
250.00	6.60E-07	6.49E-07	1.05E-06	1.08E-06
300.00	5.78E-04	5.66E-04	7.78E-04	7.92E-04
350.00	7.20E-02	7.02E-02	8.77E-02	8.88E-02
400.00	2.66E+00	2.58E+00	3.03E+00	3.06E+00
450.00	4.39E+01	4.24E+01	4.77E+01	4.80E+01
500.00	4.12E+02	3.96E+02	4.32E+02	4.34E+02

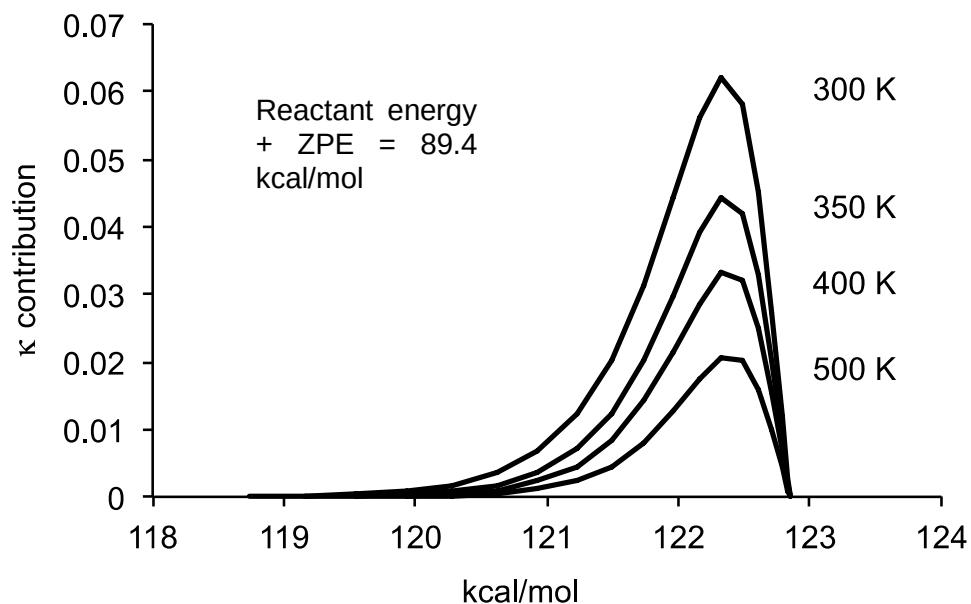
R3

Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
250.00	1.5887E+00	1.6017E+00
300.00	1.3663E+00	1.3727E+00
350.00	1.2529E+00	1.2568E+00
400.00	1.1863E+00	1.1889E+00
450.00	1.1434E+00	1.1453E+00
500.00	1.1141E+00	1.1155E+00

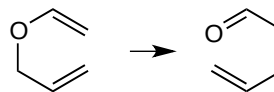
$$v_{\text{imag}}^{\ddagger} = 569.3 \text{ cm}^{-1}$$

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_{a}^{G} energy at 300-500 K.



Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
350.00	4.03E-10	4.03E-10	5.04E-10	5.10E-10
400.00	1.58E-07	1.58E-07	1.87E-07	1.88E-07
450.00	1.64E-05	1.64E-05	1.88E-05	1.89E-05
500.00	6.77E-04	6.77E-04	7.54E-04	7.58E-04

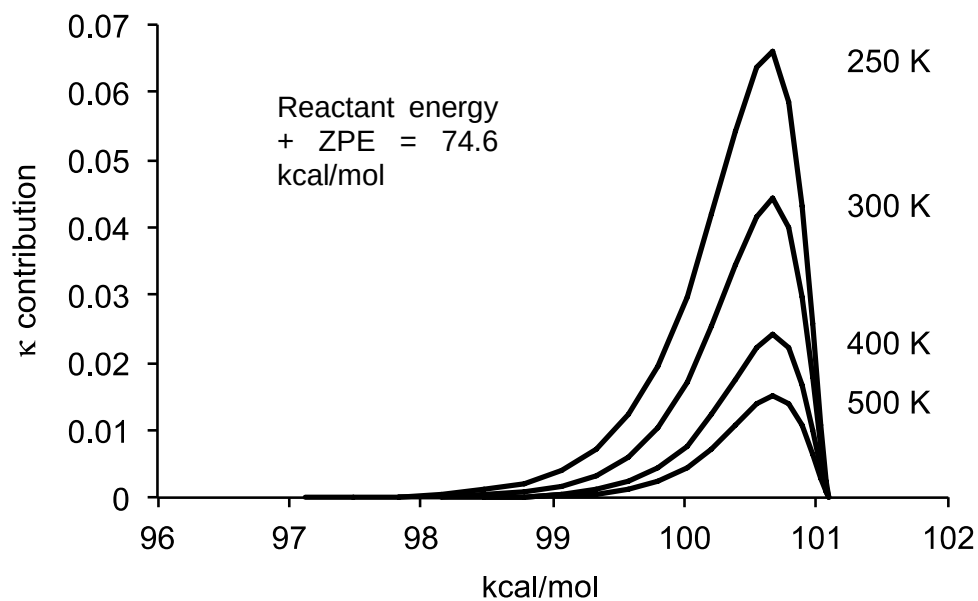
R4

Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
250.00	1.3941E+00	1.4435E+00
300.00	1.2533E+00	1.2823E+00
350.00	1.1780E+00	1.1973E+00
400.00	1.1325E+00	1.1464E+00
450.00	1.1028E+00	1.1133E+00
500.00	1.0821E+00	1.0904E+00

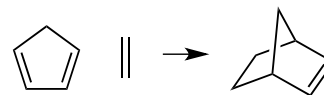
$$v_{\text{imag}}^{\ddagger} = 483.1 \text{ cm}^{-1}$$

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_{a}^{G} energy at 250-500 K.



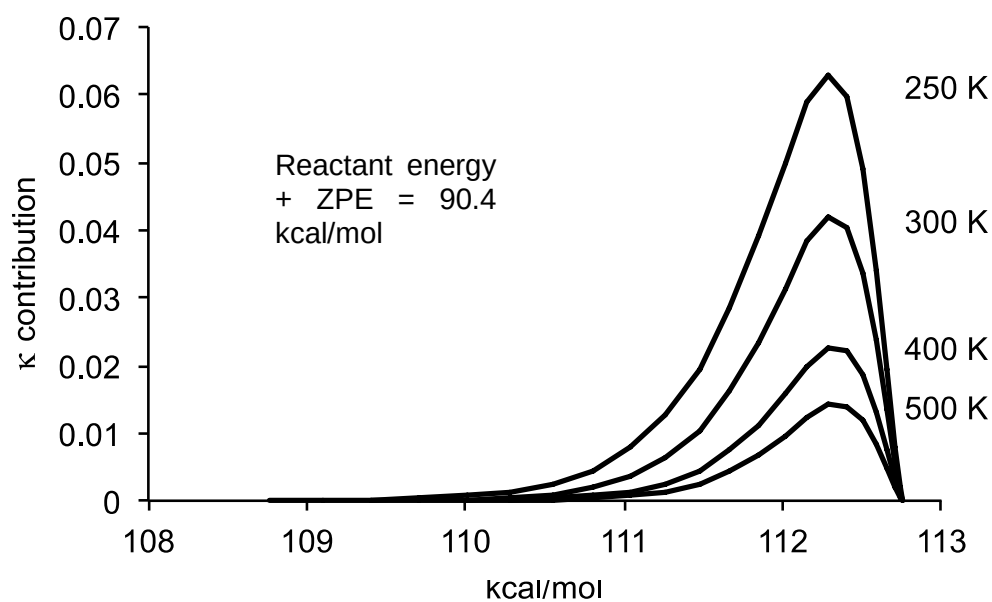
Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
250.00	1.03E-11	1.03E-11	1.44E-11	1.49E-11
300.00	7.60E-08	7.59E-08	9.51E-08	9.73E-08
350.00	4.43E-05	4.43E-05	5.22E-05	5.30E-05
400.00	5.31E-03	5.30E-03	6.01E-03	6.08E-03
450.00	2.21E-01	2.21E-01	2.43E-01	2.46E-01
500.00	4.38E+00	4.38E+00	4.74E+00	4.78E+00

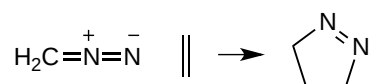
R5Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
250.00	1.4361E+00	1.4613E+00
300.00	1.2786E+00	1.2932E+00
350.00	1.1952E+00	1.2048E+00
400.00	1.1450E+00	1.1519E+00
450.00	1.1123E+00	1.1175E+00
500.00	1.0897E+00	1.0937E+00

$$v_{\text{imag}}^{\ddagger} = 469.9 \text{ cm}^{-1}$$

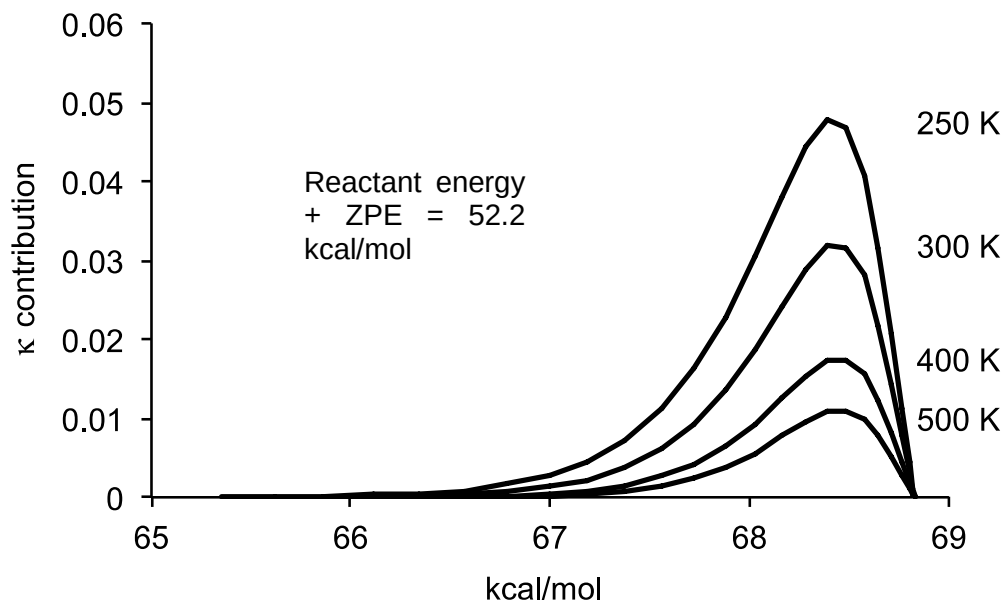
Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_{a}^{G} energy at 250-500 K.Summary of forward rate constants ($\text{cm}^3/\text{molecule-s}$) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
250.00	1.06E-35	1.05E-35	1.51E-35	1.53E-35
300.00	1.81E-32	1.79E-32	2.29E-32	2.32E-32
350.00	3.82E-30	3.79E-30	4.52E-30	4.56E-30
400.00	2.20E-28	2.18E-28	2.49E-28	2.50E-28
450.00	5.29E-27	5.23E-27	5.80E-27	5.83E-27
500.00	6.92E-26	6.83E-26	7.41E-26	7.44E-26

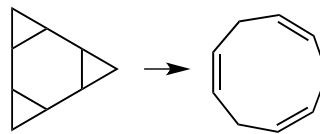
R6Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
250.00	1.3675E+00	1.3861E+00
300.00	1.2376E+00	1.2487E+00
350.00	1.1675E+00	1.1750E+00
400.00	1.1250E+00	1.1304E+00
450.00	1.0970E+00	1.1011E+00
500.00	1.0776E+00	1.0808E+00

$$v_{\text{imag}}^\ddagger = 456.8 \text{ cm}^{-1}$$

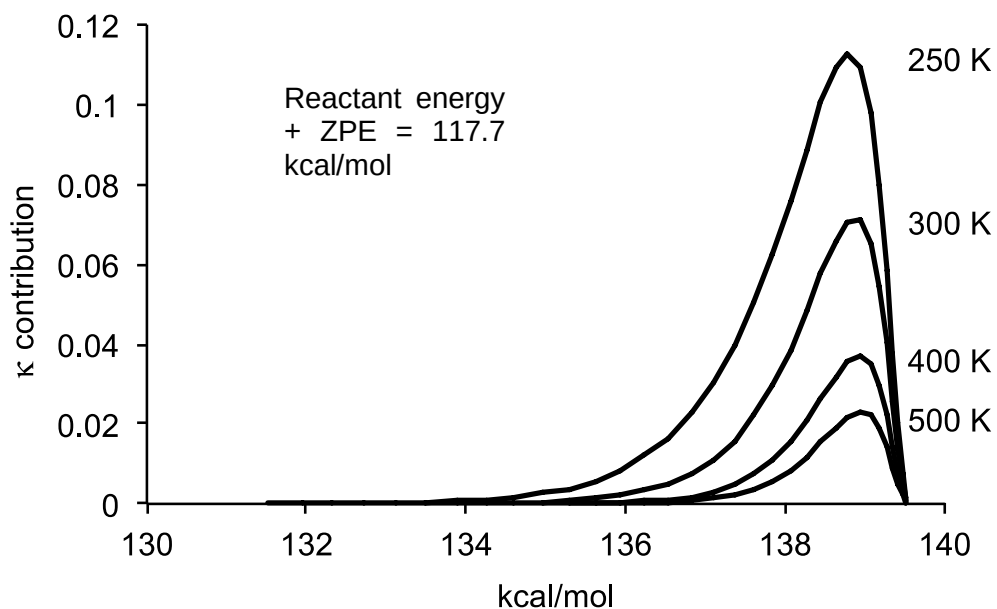
Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_a^G energy at 250–500 K.Summary of forward rate constants ($\text{cm}^3/\text{molecule-s}$) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
250.00	4.96E-29	4.78E-29	6.52E-29	6.61E-29
300.00	1.22E-26	1.18E-26	1.45E-26	1.46E-26
350.00	6.47E-25	6.19E-25	7.17E-25	7.22E-25
400.00	1.31E-23	1.25E-23	1.39E-23	1.39E-23
450.00	1.40E-22	1.33E-22	1.43E-22	1.44E-22
500.00	9.55E-22	8.99E-22	9.50E-22	9.53E-22

R7Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

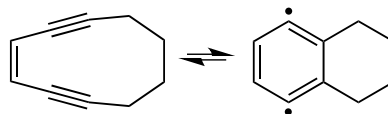
T (K)	ZCT	SCT
250.00	2.1241E+00	2.1590E+00
300.00	1.6430E+00	1.6581E+00
350.00	1.4256E+00	1.4342E+00
400.00	1.3058E+00	1.3114E+00
450.00	1.2317E+00	1.2357E+00
500.00	1.1823E+00	1.1853E+00

$$v_{\text{imag}}^{\ddagger} = 711.7 \text{ cm}^{-1}$$

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_a^{G} energy at 250-500 K.Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
250.00	5.72E-07	5.62E-07	1.19E-06	1.21E-06
300.00	1.15E-03	1.13E-03	1.86E-03	1.88E-03
350.00	2.74E-01	2.71E-01	3.86E-01	3.88E-01
400.00	1.71E+01	1.69E+01	2.20E+01	2.21E+01
450.00	4.34E+02	4.28E+02	5.27E+02	5.28E+02
500.00	5.82E+03	5.75E+03	6.79E+03	6.81E+03

R8

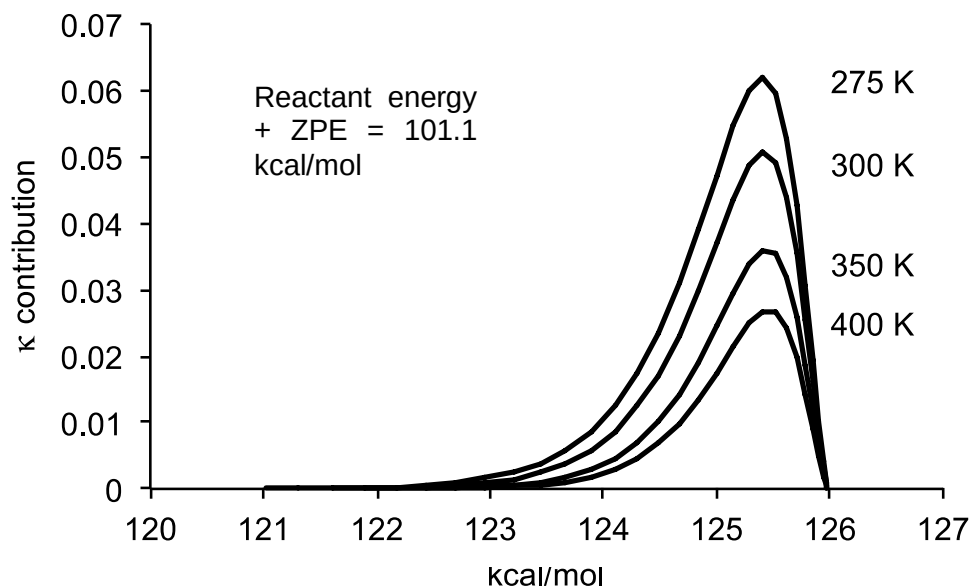


Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
250.00	1.7227E+00	1.7766E+00
275.00	1.5549E+00	1.5924E+00
300.00	1.4420E+00	1.4698E+00
350.00	1.3024E+00	1.3196E+00
400.00	1.2214E+00	1.2333E+00
500.00	1.1347E+00	1.1414E+00

$\nu_{\text{imag}}^{\ddagger} = 634.0 \text{ cm}^{-1}$
(from 10,10-CASSCF
scaled by 0.91)

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of $V_{\text{a}}^{\ddagger}$ energy at 275-400 K.



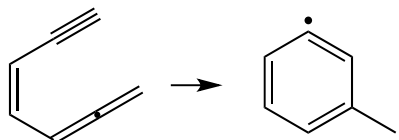
Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
275.00	2.41E-08	2.38E-08	3.62E-08	3.70E-08
300.00	1.04E-06	1.02E-06	1.44E-06	1.47E-06
310.00	3.96E-06	3.88E-06	5.33E-06	5.42E-06
350.00	3.83E-04	3.73E-04	4.72E-04	4.78E-04
400.00	3.18E-02	3.07E-02	3.62E-02	3.66E-02
500.00	1.53E+01	1.45E+01	1.57E+01	1.58E+01

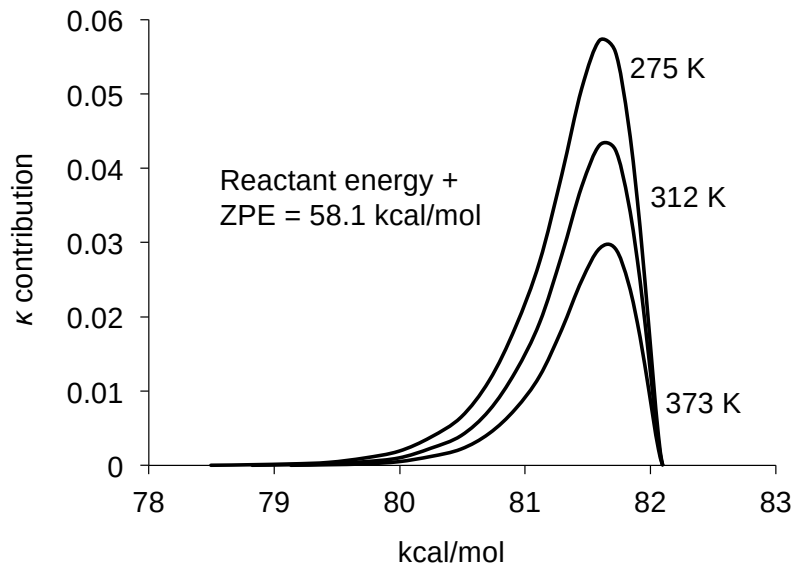
R9

Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
200.00	1.8297E+00	1.8986E+00
225.00	1.5958E+00	1.6395E+00
250.00	1.4524E+00	1.4830E+00
275.00	1.3571E+00	1.3799E+00
290.00	1.3143E+00	1.3338E+00
300.00	1.2901E+00	1.3078E+00
310.00	1.2686E+00	1.2848E+00
312.00	1.2647E+00	1.2806E+00
323.00	1.2443E+00	1.2588E+00
333.00	1.2278E+00	1.2412E+00
343.00	1.2130E+00	1.2254E+00
363.00	1.1876E+00	1.1982E+00
373.00	1.1765E+00	1.1865E+00

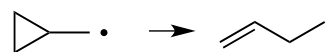


$\nu_{\text{imag}}^{\ddagger} = 632.4 \text{ cm}^{-1}$
 (from 10,10-CASSCF
 scaled by 0.91)

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of $V_a^{\ddagger}$ energy at 275–400 K.Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
200.00	1.6708E-15	1.4894E-15	2.7164E-15	2.8186E-15
225.00	1.2574E-12	1.1244E-12	1.7880E-12	1.8370E-12
250.00	2.4827E-10	2.2250E-10	3.2136E-10	3.2813E-10
275.00	1.8556E-08	1.6657E-08	2.2464E-08	2.2841E-08
290.00	1.7209E-07	1.5462E-07	2.0195E-07	2.0495E-07
300.00	6.7022E-07	6.0253E-07	7.7246E-07	7.8308E-07
310.00	2.3884E-06	2.1484E-06	2.7086E-06	2.7432E-06
312.00	3.0493E-06	2.7431E-06	3.4476E-06	3.4910E-06
323.00	1.1063E-05	9.9579E-06	1.2314E-05	1.2457E-05
333.00	3.3127E-05	2.9833E-05	3.6404E-05	3.6800E-05
343.00	9.2978E-05	8.3774E-05	1.0099E-04	1.0202E-04
363.00	6.1635E-04	5.5582E-04	6.5603E-04	6.6194E-04
373.00	1.4693E-03	1.3256E-03	1.5501E-03	1.5633E-03

R10

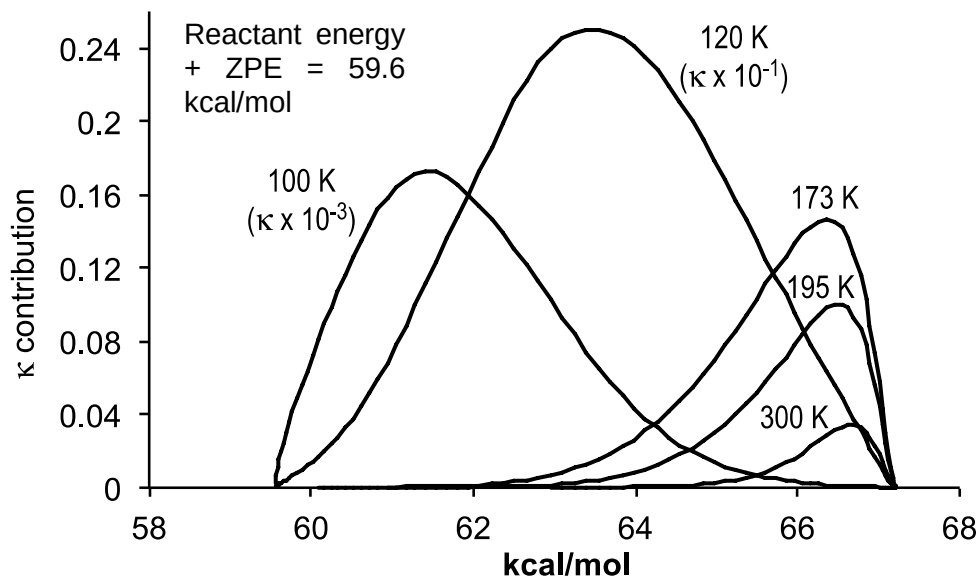


Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
100.00	1.0418E+03	4.5957E+03
120.00	3.7973E+01	7.8146E+01
173.00	3.4621E+00	4.0161E+00
195.00	2.5262E+00	2.7933E+00
250.00	1.6932E+00	1.7811E+00
273.00	1.5446E+00	1.6088E+00
295.00	1.4451E+00	1.4948E+00
300.00	1.4265E+00	1.4736E+00
350.00	1.2926E+00	1.3225E+00
353.00	1.2867E+00	1.3159E+00
400.00	1.2146E+00	1.2355E+00

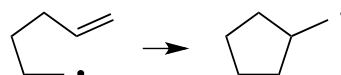
$$v_{\text{imag}}^{\ddagger} = 569.1 \text{ cm}^{-1}$$

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of $V_a^{\ddagger}$ energy at 100-300 K.



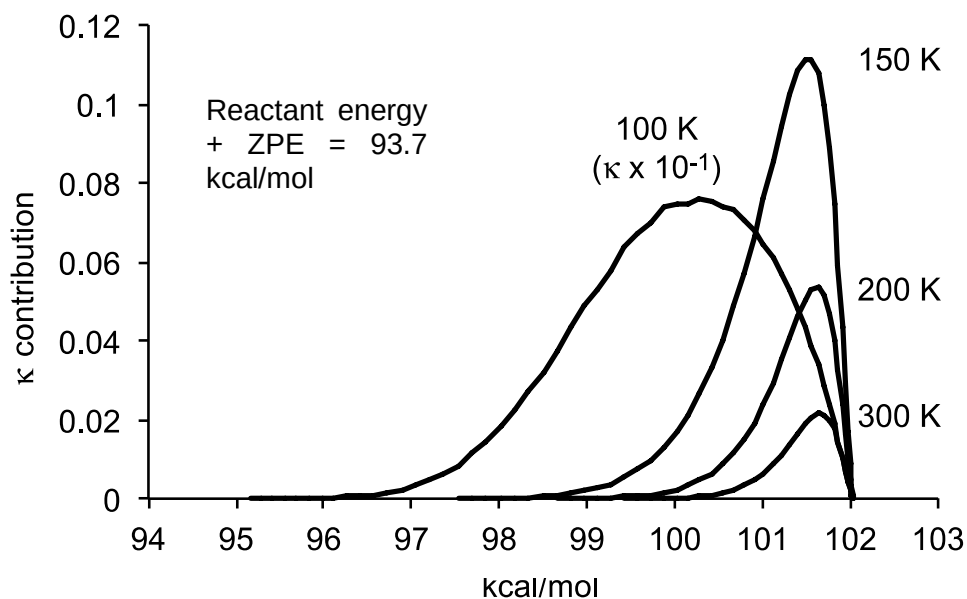
Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
100.00	3.87E-05	3.84E-05	4.00E-02	1.77E-01
120.00	2.67E-02	2.65E-02	1.01E+00	2.07E+00
173.00	6.08E+02	6.05E+02	2.09E+03	2.43E+03
195.00	7.96E+03	7.92E+03	2.00E+04	2.21E+04
250.00	6.92E+05	6.88E+05	1.17E+06	1.23E+06
273.00	2.64E+06	2.63E+06	4.06E+06	4.22E+06
295.00	7.84E+06	7.80E+06	1.13E+07	1.17E+07
300.00	9.82E+06	9.77E+06	1.39E+07	1.44E+07
350.00	6.60E+07	6.57E+07	8.48E+07	8.68E+07
353.00	7.27E+07	7.24E+07	9.31E+07	9.52E+07
400.00	2.77E+08	2.76E+08	3.35E+08	3.41E+08

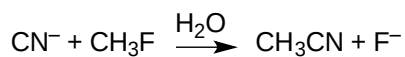
R11Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
100.00	1.2451E+01	1.8659E+01
150.00	2.3539E+00	2.5801E+00
200.00	1.5663E+00	1.6349E+00
250.00	1.3217E+00	1.3557E+00
300.00	1.2102E+00	1.2309E+00
350.00	1.1491E+00	1.1632E+00
400.00	1.1116E+00	1.1219E+00

$$v_{\text{imag}}^{\ddagger} = 436.1 \text{ cm}^{-1}$$

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_a^{G} energy at 100–300 K.Summary of forward rate constants (s⁻¹) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
100.00	4.65E-07	4.58E-07	5.70E-06	8.54E-06
150.00	5.38E-01	5.30E-01	1.24E+00	1.36E+00
200.00	5.26E+02	5.17E+02	8.06E+02	8.41E+02
250.00	3.11E+04	3.04E+04	3.99E+04	4.10E+04
300.00	4.59E+05	4.47E+05	5.35E+05	5.44E+05
350.00	3.09E+06	2.99E+06	3.39E+06	3.43E+06
400.00	1.28E+07	1.23E+07	1.35E+07	1.36E+07

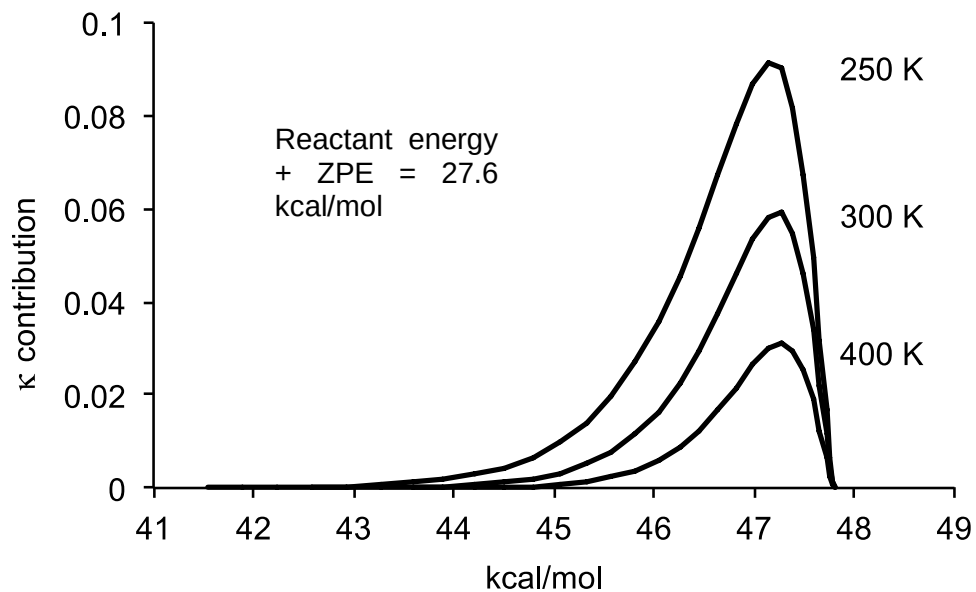
R12

Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

$$v_{\text{imag}}^\ddagger = 619.1 \text{ cm}^{-1}$$

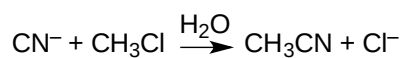
T (K)	ZCT	SCT
250.00	1.8665E+00	1.8973E+00
300.00	1.5145E+00	1.5298E+00
350.00	1.3468E+00	1.3562E+00
400.00	1.2518E+00	1.2582E+00
450.00	1.1921E+00	1.1968E+00
500.00	1.1518E+00	1.1554E+00

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_a^{G} energy at 250-400 K.



Summary of forward rate constants ($\text{cm}^3/\text{molecule-s}$) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
250.00	5.44E-31	5.13E-31	9.38E-31	9.54E-31
300.00	5.75E-28	5.39E-28	7.94E-28	8.02E-28
350.00	8.89E-26	8.27E-26	1.08E-25	1.09E-25
400.00	4.11E-24	3.80E-24	4.58E-24	4.60E-24
450.00	8.47E-23	7.76E-23	8.88E-23	8.91E-23
500.00	9.85E-22	8.96E-22	9.71E-22	9.74E-22

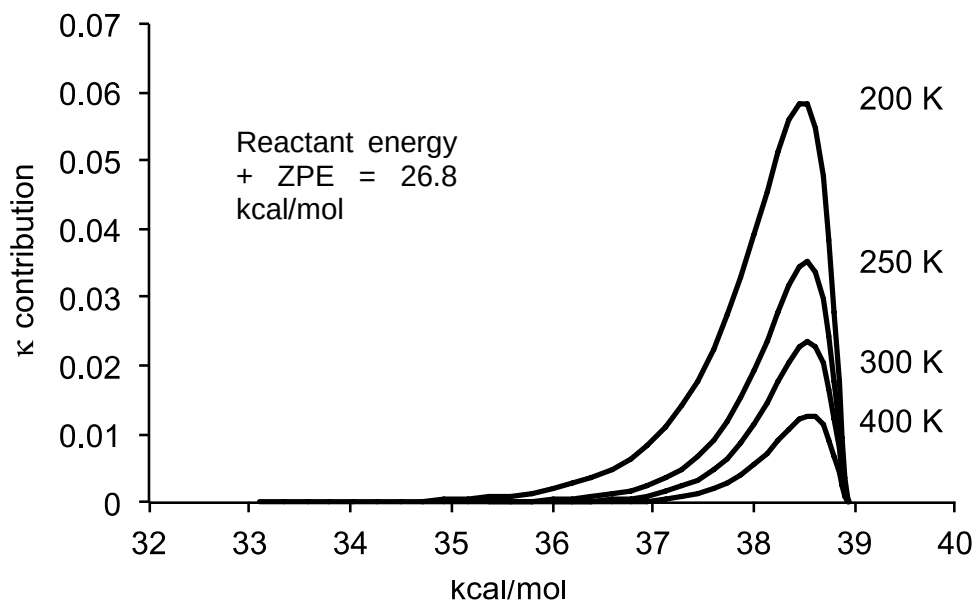
R13

Transmission coefficients $\kappa_{\text{ZCT}}(T)$ and $\kappa_{\text{SCT}}(T)$

T (K)	ZCT	SCT
200.00	1.6414E+00	1.6666E+00
250.00	1.3450E+00	1.3560E+00
300.00	1.2204E+00	1.2268E+00
350.00	1.1544E+00	1.1587E+00
400.00	1.1148E+00	1.1178E+00
450.00	1.0889E+00	1.0912E+00
500.00	1.0710E+00	1.0728E+00

$$v_{\text{imag}}^\ddagger = 501.8 \text{ cm}^{-1}$$

Contribution to $\kappa_{\text{SCT}}(T)$ as a function of V_a^G energy at 200-400 K.



Summary of forward rate constants ($\text{cm}^3/\text{molecule-s}$) :

T (K)	TST	CVT	CVT/ZCT	CVT/SCT
200.00	1.29E-26	1.11E-26	1.76E-26	1.78E-26
250.00	7.62E-24	6.39E-24	8.17E-24	8.24E-24
300.00	5.86E-22	4.82E-22	5.48E-22	5.51E-22
350.00	1.40E-20	1.12E-20	1.19E-20	1.19E-20
400.00	1.59E-19	1.25E-19	1.26E-19	1.26E-19
450.00	1.09E-18	8.42E-19	8.18E-19	8.20E-19
500.00	5.29E-18	4.00E-18	3.77E-18	3.78E-18

Full Gaussian reference 12

Gaussian 09, Revision D.01, M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery, Jr., J. E. Peralta, F. Ogliaro, M. Bearpark, J. J. Heyd, E. Brothers, K. N. Kudin, V. N. Staroverov, T. Keith, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J. M. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, O. Farkas, J. B. Foresman, J. V. Ortiz, J. Cioslowski, and D. J. Fox, Gaussian, Inc., Wallingford CT, 2013.